

SELARAS

PRODUCT REQUIREMENTS DOCUMENT

Sistem Evaluasi Layanan & Rekam Administrasi untuk Risiko Klaim

TAGLINE

Setiap pelayanan meninggalkan bukti sebelum menjadi klaim.

Healthkathon 2026 | Kategori: Efisiensi Risiko pada Fasilitas Kesehatan

Item	Value
Dokumen	Product Requirements Document
Versi	v1.0 - Baseline
Status	Ready for product build
MVP Hero	Fisioterapi - Phantom & Repeat Billing
Primary User	Reviewer / Risk & Claim Investigator
Supporting Users	Faskes Operator, Provider, Admin
Data	Synthetic / authorized data only for prototype
Architecture	Next.js + FastAPI + PostgreSQL + Neo4j + LLM

Catatan: dokumen ini adalah PRD produk, bukan proposal 20-slide. Proposal dan PPT menggunakan PRD ini sebagai source of truth.

Daftar Isi

1. Ringkasan Produk
2. Konteks dan Masalah
3. Tujuan Produk
4. Sasaran Pengguna
5. Prinsip Produk
6. Scope dan Non-Scope
7. Product Architecture
8. Functional Requirements
9. User Flows
10. UX/UI Requirements
11. Service Passport
12. Evidence & Graph Model
13. Claim Reconciliation
14. AI & Intelligence Layer
15. Risk & Review Workflow
16. Data Model
17. API & Integration
18. Tech Stack
19. Security, Privacy & Governance
20. Analytics & KPI
21. Synthetic Benchmark & Validation
22. Roadmap
23. Release Plan
24. Product Backlog
25. Acceptance Criteria
26. Definition of Done
27. Demo Specification
28. Future Expansion
29. Risks & Mitigation
30. Open Decisions

SOURCE BASIS

PRD ini mengikuti konsep SELARAS yang telah dibahas sebelumnya dan disusun agar selaras dengan struktur, batasan, dan kriteria resmi proposal Healthkathon 2026. Panduan menetapkan komponen wajib: positioning, masalah, solusi, teknis/data, maturity/prototype, rencana, dampak, governance, dan profil tim; maksimal 20 slide untuk proposal; serta data JKN riil hanya dengan izin resmi.

1. Ringkasan Produk

PRODUCT THESIS

SELARAS membentuk bukti pelayanan sejak layanan terjadi, menyatukan bukti tersebut dalam Service Passport, lalu merekonstruksi episode pelayanan untuk memastikan representasinya tetap selaras dengan billing dan klaim.

1.1 Core value proposition

- Bukti dibentuk lebih awal: bukan menunggu klaim bermasalah lalu mencari data.
- Satu episode pelayanan memiliki satu Service Passport yang dapat ditelusuri.
- Reviewer melihat cerita pelayanan, bukan hanya risk score.
- AI menjelaskan evidence gap dan memprioritaskan review; manusia tetap mengambil keputusan akhir.
- Engine dapat diperluas dengan Service Fingerprint untuk berbagai jenis layanan.

1.2 Core loop

SERVICE -> CAPTURE -> SERVICE PASSPORT -> BILLING -> CLAIM -> RECONCILE -> EXPLAIN -> HUMAN REVIEW

1.3 Hero problem

Phantom / unsubstantiated service pada layanan fisioterapi: klaim pelayanan yang tidak memiliki rangkaian bukti pelayanan yang cukup, lengkap, atau konsisten dari data yang tersedia.

2. Konteks dan Masalah

2.1 Problem statement

Dalam satu episode pelayanan, bukti tersebar pada appointment, arrival/encounter, provider assignment, treatment event, clinical note, completion, billing, dan claim. Ketika reviewer menerima klaim, hubungan antarevent tersebut tidak selalu terlihat sebagai satu cerita yang utuh.

SELARAS menjawab gap tersebut dengan mengubah jejak pelayanan menjadi evidence chain yang dapat direkonstruksi dan dibandingkan dengan representasi klaim.

2.2 Why now

- Volume data pelayanan dan klaim membuat pemeriksaan satu per satu semakin tidak efisien.
- Risk signal tanpa evidence context dapat menyulitkan reviewer memahami apa yang harus diperiksa.
- Kebutuhan interoperabilitas RME dan digital health membuat struktur evidence lintas sumber semakin relevan.
- Healthkathon 2026 secara eksplisit menempatkan phantom/repeat billing dan risiko fasilitas kesehatan lain sebagai challenge.

2.3 User pain

Pain	Current effect	SELARAS response
Evidence tersebar	Reviewer mencari lintas sumber	Service Passport
Klaim ringkas	Konteks episode hilang	Episode Reconstruction
Anomaly sulit dijelaskan	Reviewer perlu investigasi manual	AI Evidence Reasoner
Dampak tidak terlihat	Sulit memahami materiality	Claim Impact / Counterfactual
Risk ≠ fraud	False positive berisiko	Human-in-the-loop

3. Tujuan Produk

3.1 Primary goals

- Mengurangi waktu yang dibutuhkan reviewer untuk merekonstruksi satu episode pelayanan.
- Meningkatkan keterlacakan antara service evidence, billing, dan claim.
- Mengidentifikasi evidence gap dan contradiction secara explainable.
- Membentuk Service Passport sejak point-of-care untuk layanan pilot.
- Menguji reconciliation engine pada synthetic benchmark dengan ground truth yang terkontrol.

3.2 Secondary goals

- Menyediakan fondasi Service Fingerprint untuk multi-service.
- Menyediakan data provenance dan audit log sejak awal.
- Menyediakan arsitektur yang siap diintegrasikan dengan sistem existing tanpa menggantikannya.

3.3 Non-goals

- Menentukan fraud secara otomatis.
- Menggantikan RME/SIMRS/SATUSEHAT/claim system.
- Menjadi aplikasi klinis atau alat diagnosis.
- Membuat clinical outcome score sebagai sumber keputusan utama.
- Menggunakan data peserta JKN riil tanpa otorisasi resmi.

4. Sasaran Pengguna

Persona	Need	Primary screens	Success signal
Faskes Operator	Memulai dan menyelesaikan layanan dengan bukti minimal	Live Service, Passport	Episode terbentuk tanpa field wajib kosong
Tenaga Kesehatan	Konfirmasi treatment dan clinical note	Service Detail	Treatment + note terkait episode
Reviewer	Memahami claim dan memutuskan tindak lanjut	Claim Queue, Replay, Graph, Review	Review time turun, alasan jelas
Admin	Mengatur user, template, audit	Admin, Service Template, Audit	Perubahan terkontrol dan terlacak

4.1 Primary persona: Reviewer

JOB TO BE DONE

“Ketika sebuah klaim perlu diperiksa, bantu saya memahami apa yang diklaim, apa yang benar-benar memiliki bukti, di mana mismatch terjadi, dan apa yang harus saya periksa lebih dahulu.”

5. Prinsip Produk

Principle	Meaning
Evidence first	Mulai dari kejadian pelayanan dan evidence, bukan dari label fraud.
Missing evidence is not fraud	Evidence gap menghasilkan Need Review, bukan vonis.
Explain before score	Setiap risk signal harus memiliki alasan dan evidence trail.
Human decides	AI memberi rekomendasi; reviewer membuat keputusan final.
Point-of-care by design	Evidence dibentuk sedekat mungkin dengan saat pelayanan terjadi.
Synthetic-first for prototype	Semua uji kompetisi menggunakan data sintetis/berizin.
Modular service fingerprint	Perluasan layanan dilakukan melalui template/footprint.
Traceable by default	Setiap evidence memiliki source, event, timestamp, dan provenance.

6. Scope dan Non-Scope

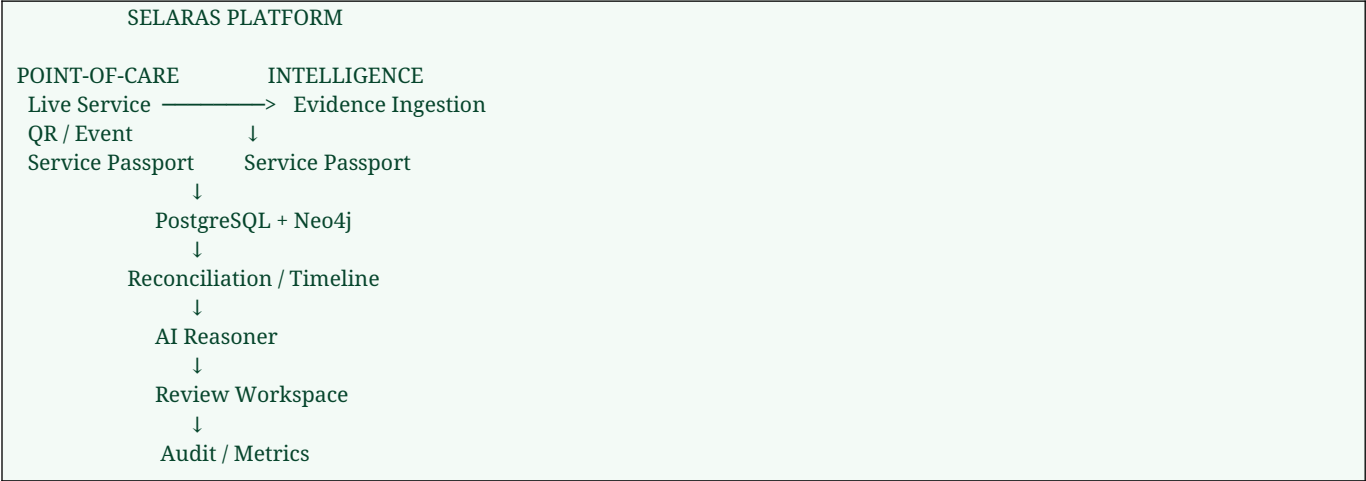
6.1 MVP in-scope

- Service Capture untuk fisioterapi.
- Service Passport.
- Evidence Chain.
- Expected Service Footprint.
- Observed Service Footprint.
- Claim-Service Reconciliation.
- Evidence Coverage & Proof Gap.
- Temporal consistency checks dasar.
- Claim Replay.
- Evidence Graph berbasis Neo4j.
- AI Evidence Reasoner.
- Human Review Workspace.
- Basic Claim Impact simulation.
- Audit Log dan role-based access.

6.2 Out-of-scope MVP

- Production SATUSEHAT integration.
- Clinical diagnosis or treatment recommendation.
- Automated fraud determination.
- Real participant data ingestion without authorization.
- Complex ML fraud model trained on unknown real-world labels.
- Full multi-service rollout.

7. Product Architecture



7.1 Architectural principles

- Transactional source of truth in PostgreSQL.
- Relationship and traversal layer in Neo4j.
- LLM receives curated structured evidence, not uncontrolled raw PHI.
- Evidence provenance is immutable at the application layer where possible.
- Frontend and backend are independently deployable.

8. Functional Requirements

ID	Requirement	Priority	Acceptance summary
FR-01	Authentication & RBAC	P0	User login and role-based access for Admin, Provider, Operator, Reviewer.
FR-02	Live Service	P0	Create/start/end a service episode with patient, provider, service, time, service point.
FR-03	Service Passport	P0	Create a passport for every service episode with evidence status.
FR-04	Evidence Ingestion	P0	Attach structured evidence events and provenance.
FR-05	Service Fingerprint	P0	Define expected evidence template for physiotherapy.
FR-06	Claim Ingestion	P0	Create/import synthetic claim and claim items.
FR-07	Reconciliation	P0	Compare expected/observed service footprint with claim representation.
FR-08	Evidence Coverage	P0	Calculate coverage and identify proof gaps.
FR-09	Replay	P0	Render ordered episode timeline with evidence details.
FR-10	Evidence Graph	P0	Visualize episode relationship graph in Neo4j.
FR-11	AI Explanation	P1	Generate grounded explanation of mismatch and review rationale.
FR-12	Review Workspace	P0	Allow human decision and reviewer notes.
FR-13	Claim Impact Simulation	P1	Show synthetic counterfactual/potential impact.
FR-14	Audit Log	P0	Log key user/system actions.
FR-15	Analytics	P1	Evidence coverage, review time, queue and benchmark metrics.

9. User Flows

9.1 Flow A - Start service

Login -> Select patient -> Select service -> Select provider -> Confirm service point -> Start -> Event created -> Passport opened

9.2 Flow B - Complete service

Passport open -> Add treatment -> Add note -> Complete service -> Evidence chain update -> Coverage recalculated -> Passport status updated

9.3 Flow C - Review claim

Claim Queue -> Open Claim -> View coverage -> Select flagged item -> Replay -> Evidence Graph -> AI Explanation -> Review Decision -> Audit Log

9.4 Flow D - Counterfactual

Claim Detail -> View Potential Impact -> Select “simulate without mutation” -> Recalculate synthetic scenario -> Show difference -> Save as non-decision insight

10. UX/UI Requirements

10.1 Design direction

- Enterprise health-tech: clean, calm, evidence-centric.
- Primary color: deep forest green; supporting emerald, warm white, neutral gray; amber/red only for state signals.
- Dense enough for reviewer productivity, but not visually noisy.
- One screen = one decision question.
- No “AI magic” animations that obscure evidence.

10.2 Information architecture

Sidebar

- Overview
- Pelayanan
- Service Passport
- Claim Queue
- Replay
- Evidence Graph
- Analytics
- Service Templates
- Audit Log
- Settings

10.3 Landing page sections

- Hero: “Setiap pelayanan meninggalkan bukti sebelum menjadi klaim.”
- Problem / gap: Claim vs service evidence.
- How it works: Capture -> Passport -> Reconstruct -> Reconcile.
- Interactive golden case.
- AI Evidence Reasoner.
- Impact / metrics.
- Service Fingerprint scalability.
- Privacy & governance.

11. Service Passport

11.1 Concept

Service Passport adalah objek produk inti yang merepresentasikan satu episode pelayanan. Passport menyatukan identitas episode, evidence, timeline, coverage, provenance, dan status review.

SERVICE PASSPORT
ID: SRV-1025-08
Patient: P-1025 (synthetic)
Service: Physiotherapy
Provider: T-031
Start: 14:03
End: 14:41

Evidence
✓ Identity
✓ Provider
✓ Service Point
✓ Start Time
✓ Treatment
✓ Clinical Note
✓ Completion

Coverage: 100%
Status: SUPPORTED

11.2 Passport states

State	Meaning
DRAFT	Episode dibuat, service belum dimulai.
ACTIVE	Pelayanan sedang berlangsung.
COMPLETE	Pelayanan selesai dan evidence minimum ada.
INCOMPLETE	Ada expected evidence yang belum ditemukan.
REVIEW	Episode/claim terkait perlu diperiksa.
VERIFIED	Reviewer menutup kasus setelah verifikasi.

12. Evidence & Graph Model

12.1 Neo4j model

```
(:Patient)-[:HAS_EPISODE]->(:ServiceEpisode)
(:ServiceEpisode)-[:HAS_EVENT]->(:ServiceEvent)
(:ServiceEvent)-[:SUPPORTED_BY]->(:Evidence)
(:ServiceEpisode)-[:HAS_PASSPORT]->(:ServicePassport)
(:ServiceEpisode)-[:GENERATES]->(:Billing)
(:Billing)-[:REFERENCED_BY]->(:Claim)
(:Claim)-[:CLAIMS]->(:ClaimItem)
(:ClaimItem)-[:MAPS_TO]->(:ServiceEpisode)
```

12.2 Evidence types

Evidence type	Example	Trust/use
Operational	Arrival, provider assignment, treatment event	Core corroboration
Clinical	Clinical note, assessment	Supporting clinical evidence
Administrative	Billing, claim item	Claim-side evidence
Device/service signal	Session/device activity	Supporting signal
Derived AI	AI explanation / embedding	Interpretation only; not source evidence

12.3 Provenance fields

```
evidence_id
source_system
source_record_id
recorded_at
effective_at
actor_id
created_at
data_version
confidence / quality flag
```

13. Claim Reconciliation

13.1 Expected vs observed vs claimed



13.2 Core calculations

Metric	Definition	MVP behavior
Evidence Coverage	Supported required evidence / expected evidence	Percentage per episode/claim
Proof Gap	Expected evidence not sufficiently supported	List of missing/weak evidence
Quantity Divergence	Claimed units - supported units	Flag if non-zero beyond rule
Temporal Issue	Events violate expected order/window	Flag for review
Evidence Contradiction	Two sources disagree materially	Higher review priority after validation

13.3 Important rule

SAFETY RULE

Evidence gap is a review signal, not a fraud verdict. A legitimate data synchronization or documentation issue must remain possible until a human reviewer verifies the case.

14. AI & Intelligence Layer

14.1 AI components

Component	Purpose	MVP status
Episode Reconstruction	Turn structured evidence into a readable episode story	P1
Temporal Context	Summarize timing and sequence	P0/P1
Evidence Consistency	Assess whether observed evidence supports expected footprint	P0 rule engine + P1 AI
Semantic Mutation	Detect meaningful changes in diagnosis/procedure/coding	P2
Claim Impact	Estimate synthetic impact	P1
Counterfactual	Simulate without selected mutation	P1
AI Evidence Reasoner	Explain what/when/evidence/gap/why review	P1
Risk Prioritization	Sort cases using transparent features	P1

14.2 AI grounding contract

- AI explanation must cite internal evidence IDs/timestamps.
- LLM cannot invent evidence that is absent from the structured context.
- No direct “fraud = true/false” output.
- Low-confidence explanation must be labeled as such.
- All final decisions remain with reviewer.

15. Risk & Review Workflow



15.1 Review card

- Claim summary
- Evidence coverage
- Proof gap
- Timeline
- Graph
- AI explanation
- Potential impact
- Reviewer decision + notes

16. Data Model

16.1 PostgreSQL entities

Entity	Key fields	Purpose
users	id, role, facility_id	Authentication and RBAC
patients	id, synthetic_key	Synthetic identity
providers	id, role	Provider identity
service_templates	id, service_type, required_evidence	Expected footprint
service_episodes	id, patient_id, service_type, provider_id, status	Core episode
service_events	id, episode_id, event_type, occurred_at	Structured events
evidence	id, episode_id, type, source, occurred_at, quality	Evidence objects
service_passports	id, episode_id, coverage, status	Passport snapshot
billings	id, episode_id, amount, created_at	Billing-side data
claims	id, claim_ref, facility_id	Claim header
claim_items	id, claim_id, service_type, quantity, amount	Claim units
reviews	id, claim_id, reviewer_id, decision, note	Human outcome
audit_logs	id, actor, action, target, created_at	Audit trail

16.2 Neo4j nodes

Patient ServiceEpisode ServicePassport ServiceEvent Evidence Billing Claim ClaimItem Review

17. API & Integration

17.1 Core endpoints

Method	Endpoint	Purpose
POST	/api/v1/auth/login	Login
POST	/api/v1/service-episodes	Create episode
POST	/api/v1/service-episodes/:id/start	Start service
POST	/api/v1/service-episodes/:id/events	Add event/evidence
POST	/api/v1/service-episodes/:id/complete	Complete service
GET	/api/v1/service-passports/:id	Passport detail
POST	/api/v1/claims	Create/import claim
GET	/api/v1/claims/queue	Review queue
GET	/api/v1/claims/:id/replay	Episode replay
GET	/api/v1/claims/:id/graph	Graph payload
POST	/api/v1/claims/:id/reconcile	Run reconciliation
POST	/api/v1/claims/:id/explain	Generate grounded AI explanation
POST	/api/v1/claims/:id/simulate	Counterfactual simulation
POST	/api/v1/reviews	Submit reviewer decision
GET	/api/v1/audit-logs	Audit logs

17.2 Integration stance

- MVP exposes REST APIs and mock import/export.
- Future interoperability should align to authorized health data standards and institutional interfaces.
- Never claim production SATUSEHAT integration without authorization and testing.

18. Tech Stack

Layer	Technology	Why
Web	Next.js 16 + TypeScript	Fast product iteration, App Router, strong UI ecosystem
UI	Tailwind CSS + shadcn/ui	Consistent enterprise interface
Animation	Framer Motion	Controlled micro-interactions
Backend	FastAPI + Python	Data/AI/graph processing friendly
Primary DB	PostgreSQL	Transactional source of truth
Graph DB	Neo4j	Episode/evidence relationships and traversal
Cache/Queue	Redis	Async processing and caching
Auth	Supabase Auth or JWT	Fast RBAC setup for prototype
Object Storage	Supabase Storage / S3-compatible	Evidence attachments
AI	LLM API + rule engine	Grounded explanation + deterministic checks
Testing	Pytest + Vitest / Playwright	Backend, frontend, E2E
Observability	Sentry + structured logs	Error tracking and diagnostics
Deployment	Vercel + Railway/Render + managed Postgres/Neo4j	Prototype-friendly deployment

ARCHITECTURE DECISION

Neo4j dipakai karena Evidence Graph adalah bagian inti dari product experience dan reviewer membutuhkan traversal episode/evidence. PostgreSQL tetap menjadi transactional source of truth agar workflow bisnis tidak bergantung pada graph database semata.

19. Security, Privacy & Governance

Area	Requirement	Implementation direction
Privacy	Minimize sensitive data	Synthetic patient IDs for prototype; field-level minimization
Access	RBAC	Admin / Operator / Provider / Reviewer / Auditor
Transport	Encryption in transit	HTTPS/TLS
Storage	Encryption at rest	Managed DB/storage encryption
Audit	Immutable-ish audit trail	Append-only application audit log
Provenance	Source traceability	source_system + source_record_id + timestamps
AI	Human decision	AI output is advisory
Bias	Scenario testing	Synthetic cases including benign changes
Prompt safety	Grounded context only	Structured evidence context, no uncontrolled PHI dump
Retention	Configurable retention	Policy-driven lifecycle

19.1 AI failure handling

- If LLM unavailable: display deterministic evidence summary.
- If evidence incomplete: show gap; do not fabricate a narrative.
- If AI confidence low: label low confidence and require reviewer inspection.
- If source conflict: surface the conflict explicitly instead of selecting one silently.

20. Analytics & KPI

KPI	Definition	Measurement stage	Target type
Evidence Coverage	Supported evidence / expected evidence	Prototype + pilot	Baseline + improve
Review Time	Time from case open to reviewer decision	Pilot	Reduce
Reconstruction Time	Time to obtain complete episode view	Prototype + pilot	Reduce
Proof Gap Detection	Known benchmark gaps detected	Synthetic benchmark	Measure recall/precision
Review Prioritization	High-risk benchmark items surfaced earlier	Synthetic benchmark	Measure ranking quality
AI Explanation Grounding	Statements trace to evidence IDs	Prototype	100% for cited claims
System Reliability	Successful transaction rate	Prototype	Measure SLA target

MEASUREMENT RULE

Angka performa yang belum diuji tidak boleh ditulis sebagai hasil. Gunakan label “target”, “benchmark result”, “prototype result”, atau “pilot result” sesuai status bukti.

21. Synthetic Benchmark & Validation

21.1 Dataset design

Scenario	Expected behavior	Ground truth
Normal complete	Clear / supported	No intentional issue
Missing treatment	Need review	Treatment event absent
Missing note	Need review	Clinical note absent
Duplicate service	Review	Same service mapped twice
Quantity mismatch	Review	Claim quantity > supported quantity
Temporal inconsistency	Review	Expected sequence violated
Benign late documentation	Do not over-escalate	Late note with legitimate evidence
Legitimate diagnosis update	Do not flag solely on change	New supporting test/evidence exists

21.2 Validation procedure

- 31. Generate synthetic episodes and claims.
- 32. Insert controlled risk patterns.
- 33. Run reconciliation engine.
- 34. Compare results to ground truth.
- 35. Inspect false positives and false negatives.
- 36. Review AI explanation against evidence source IDs.
- 37. Repeat after rule/model changes.

22. Roadmap

Phase	Timeline	Focus	Key deliverables
P0 Foundation	Weeks 1-2	Architecture + data	Repo, auth, schema, seed data
P1 Service Capture	Weeks 3-4	Point-of-care	Live Service, events, passport
P2 Reconciliation	Weeks 5-6	Claim intelligence	Footprint, coverage, gap, queue
P3 Explain	Weeks 7-8	Replay + graph + AI	Replay, Neo4j, explanation, review
P4 Hardening	Weeks 9-10	Quality	Tests, governance, benchmark, demo
P5 Pilot Readiness	Post-hackathon	Security + integration	RBAC, provenance, audit, connectors
P6 Multi-service	Later	Service fingerprints	Radiology, lab, pharmacy, devices
P7 Advanced AI	Later	Mutation + impact	Semantic mutation, counterfactual, pattern learning

23. Release Plan

Release 0.1 - Internal prototype

- Single facility seed data.
- Synthetic patient + claims.
- Service Capture + Passport.
- Basic reconciliation.

Release 0.2 - Demo candidate

- Golden Case end-to-end.
- Neo4j graph.
- Replay.
- AI explanation grounded to evidence.
- Review submission + audit.

Release 0.3 - Validation build

- Synthetic benchmark harness.
- Scenario generator.
- Metrics dashboard.
- Regression suite.

Release 1.0 - Pilot-ready concept

- RBAC + audit hardening.
- Data provenance controls.
- Configurable service templates.
- Operational monitoring.

24. Product Backlog

Epic	Stories	Priority
Foundation	Auth, RBAC, schema, seed, navigation	P0
Service Capture	Create/start/complete service	P0
Passport	Evidence status + coverage + provenance	P0
Claim	Claim import + queue + item mapping	P0
Reconciliation	Expected vs observed vs claimed	P0
Replay	Timeline + evidence drawer	P0
Graph	Neo4j model + visualizer	P0
Review	Decision workflow + notes	P0
AI	Evidence explanation + summary	P1
Impact	Counterfactual simulation	P1
Analytics	Benchmark + review metrics	P1
Admin	Service templates + audit	P1
Service expansion	Radiology/lab/pharmacy	P2
Advanced AI	Semantic mutation + pattern learning	P2

25. Acceptance Criteria

ID	Acceptance
AC-01	Operator can start and complete a physiotherapy episode without manual DB intervention.
AC-02	Every completed episode generates a Service Passport with status and evidence list.
AC-03	Claim can be mapped to one or more service episodes using synthetic claim items.
AC-04	System displays Evidence Coverage and Proof Gap for every mapped episode/claim.
AC-05	Replay renders ordered events by effective timestamp and distinguishes source events.
AC-06	Neo4j graph shows patient -> episode -> evidence -> billing -> claim relationships.
AC-07	AI explanation cites evidence IDs and does not assert fraud as fact.
AC-08	Reviewer can submit Verified / Need Clarification / Follow-up with note.
AC-09	All review actions appear in audit log.
AC-10	Golden Case can run end-to-end in a clean environment using seed data.
AC-11	Synthetic benchmark can be executed and metrics exported.
AC-12	System gracefully handles missing evidence and AI unavailability.

26. Definition of Done

- Feature implemented and code reviewed.
- API contract documented.
- Unit tests added for core business logic.
- E2E test added for critical workflow where feasible.
- Error and empty states implemented.
- Role access verified.
- Audit events emitted for sensitive actions.
- No hard-coded secret or API key in repository.
- No real participant data used in demo dataset.
- Documentation updated.
- Demo seed can reproduce the Golden Case deterministically.
- Visual QA passed on desktop and mobile where applicable.

27. Demo Specification

GOLDEN DEMO

Satu cerita. Satu pasien sintetis. Satu klaim. Semua kemampuan utama muncul dari episode yang sama.

Time	Moment	Action
0:00	Start Service	Scan/select patient + provider -> start
0:20	Create Passport	Evidence appears in real time
0:40	Complete Service	Passport gains treatment/note/completion
1:00	Claim Arrives	10 sessions claimed
1:20	Reconcile	8 supported, 2 need review
1:40	Replay	Show exact timeline gap
2:00	Graph	Show broken/weak evidence relationship
2:20	AI	Explain what changed, evidence, gap, why review
2:40	Impact	Show synthetic potential claim impact
3:00	Human Decision	Need Clarification + note

28. Future Expansion

28.1 Service Fingerprint framework

Physiotherapy -> Radiology -> Laboratory -> Pharmacy -> Medical Devices -> Inpatient

28.2 Future capabilities

- Computer vision/service signal as optional corroborating evidence.
- Semantic mutation detection for diagnosis/procedure/coding changes.
- Cross-episode pattern analysis.
- Network-level risk pattern visualization.
- Configurable rules per facility/service.
- Interoperability connectors after authorization.
- Pilot feedback loop for threshold calibration.

LONG-TERM PRODUCT IDEA

Satu lapisan integritas, banyak service fingerprint: engine tetap konsisten, sedangkan definisi evidence menyesuaikan karakter layanan.

29. Risks & Mitigation

Risk	Severity	Mitigation
False positive	High	Human review, explainability, benchmark benign changes
Missing/late data	High	Provenance, late-event handling, gap ≠ fraud
LLM hallucination	High	Grounded context, evidence IDs, deterministic fallback
Privacy breach	High	Synthetic-first, RBAC, encryption, least privilege
Graph complexity	Medium	Keep Postgres as source of truth; Neo4j for relationships
Scope creep	High	MVP locked to physiotherapy hero case
Integration assumptions	Medium	Mock interfaces until authorized
User adoption	Medium	Minimize capture friction and test with user scenarios
Unvalidated CV signal	Medium	Treat as optional supporting signal

30. Open Decisions

- Final authentication provider: Supabase Auth vs self-managed JWT.
- Exact Neo4j hosting choice for prototype.
- LLM provider/model and budget ceiling.
- Whether Service Signal/CV is included in the live demo or shown as future-ready integration.
- Exact synthetic claim/value schema used for counterfactual impact.
- Final service fingerprint fields for physiotherapy.
- Reviewer role taxonomy and approval workflow.
- Final hosting/domain strategy.

Appendix A - Golden Data Example

Episode	Claimed	Supported	Main issue	Expected reviewer action
EP-001	1	1	None	Verified
EP-002	1	0.7	Treatment missing	Need clarification
EP-003	2	2	Temporal inconsistency	Review timeline
EP-004	3	2	Quantity mismatch	Check claim items
EP-005	1	1	Late note only	Review as benign candidate

DATA STATUS

Semua contoh pada appendix adalah ilustrasi desain / synthetic scenarios. Jangan menyajikannya sebagai data JKN nyata atau hasil pilot.

Appendix B - Proposal Alignment Checklist

Requirement	Covered in PRD	Notes
Identitas & positioning	Yes	Sections 1 and 2
Masalah & urgensi	Yes	Section 2
Solusi & keunggulan	Yes	Sections 6-15
Pendekatan teknis & data	Yes	Sections 16-18
Maturity / prototype	Yes	Sections 22-27
Rencana & kelayakan	Yes	Sections 22-23
Dampak & nilai terukur	Yes	Section 20-21
Risiko, privacy, ethics	Yes	Section 19 and 29
Profil tim	To fill	Depends on final team and actual portfolio
One category / one solution	Yes	Facility risk / SELARAS
Synthetic data for prototype	Yes	Explicit throughout
Human-in-the-loop	Yes	Sections 14-15, 19

FINAL PRODUCT PRINCIPLE

SELARAS bukan “AI yang menuduh”. SELARAS adalah sistem yang membuat hubungan antara pelayanan, bukti, billing, dan klaim menjadi terlihat, dapat dijelaskan, dan dapat ditindaklanjuti reviewer.